

SERENITY

Smart Emotion Recognition & Neural Intervention Technology

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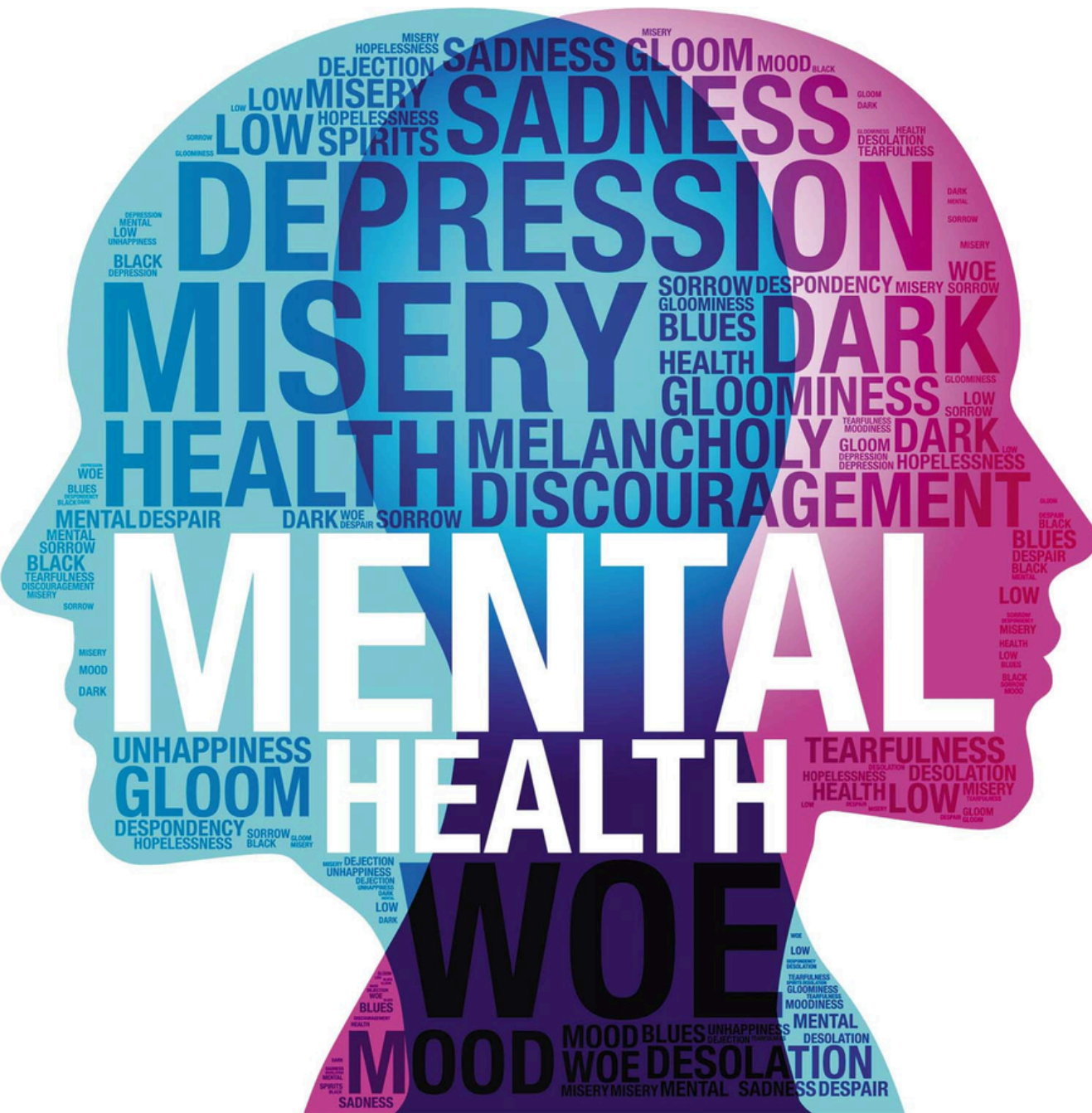
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PROBLEM & MOTIVATION



- **Problem:** Timely, low-cost mental-health support is hard to access; crises can occur anytime.
- **Why Now:** Rising post-pandemic mental-health needs + affordable edge AI (Raspberry Pi 5).
- **Engineering Context:** On-device mental health care for privacy.
- **Constraints:** <1 s latency, <10 W power, multilingual, secure & safe.

OBJECTIVES & SCOPE



Objectives

- Real-time **facial + vocal** sentimental analysis
- Deploy **multilingual**, therapy-tuned LLM on **Raspberry Pi 5**
- Personalized **long-term memory per user**
- **Edge optimization** (INT8/FP16 quant., pruning)

Scope

- **In-scope:** empathy, CBT-style dialogue, edge privacy
- **Out-of-scope:** medical diagnosis or licensed therapy

Success metrics: ≤ 1 s E2E latency, ≥ 95 % emotion accuracy, $\geq 4/5$ empathy score

LITERATURE REVIEW & RELEVANT PRODUCT SEARCH

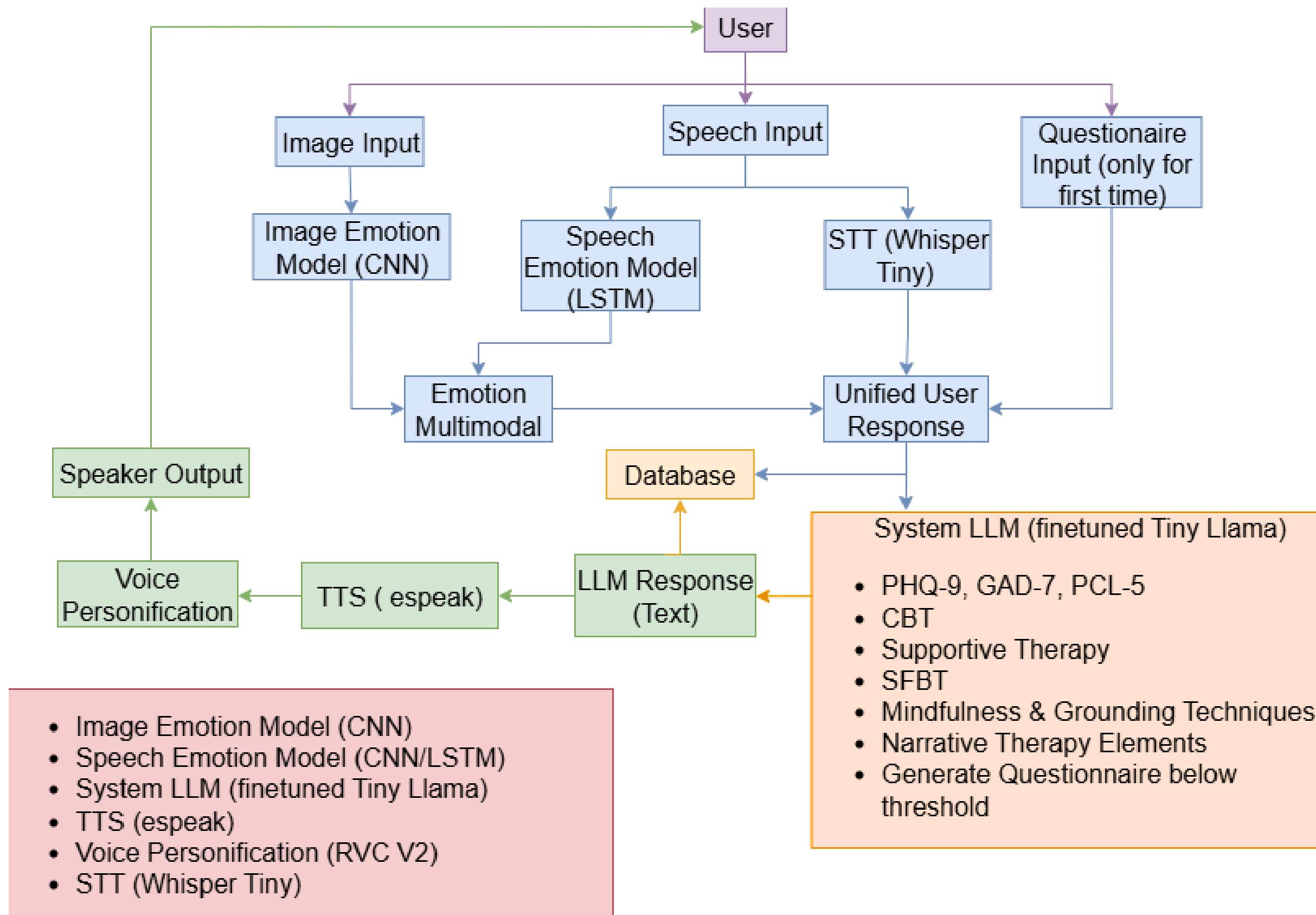
- “A Method of Mental Care Using CNN, Transformer, TTS, Voice Conversion, and Whisper”
- "Attention Is All You Need" (2017)
- Keywords: “multimodal emotion recognition,” “empathetic dialogue,” “LLM fine-tuning empathy,” “speech + facial emotion recognition.”

- Date Range: 2018 – 2025 (focused on LLM fine-tuning works).
- Addresses emotion recognition or empathetic dialogue
- Report clear metrics (accuracy, latency, empathy scores)
- Practical deployment relevance

KEY PAPERS & PRODUCTS

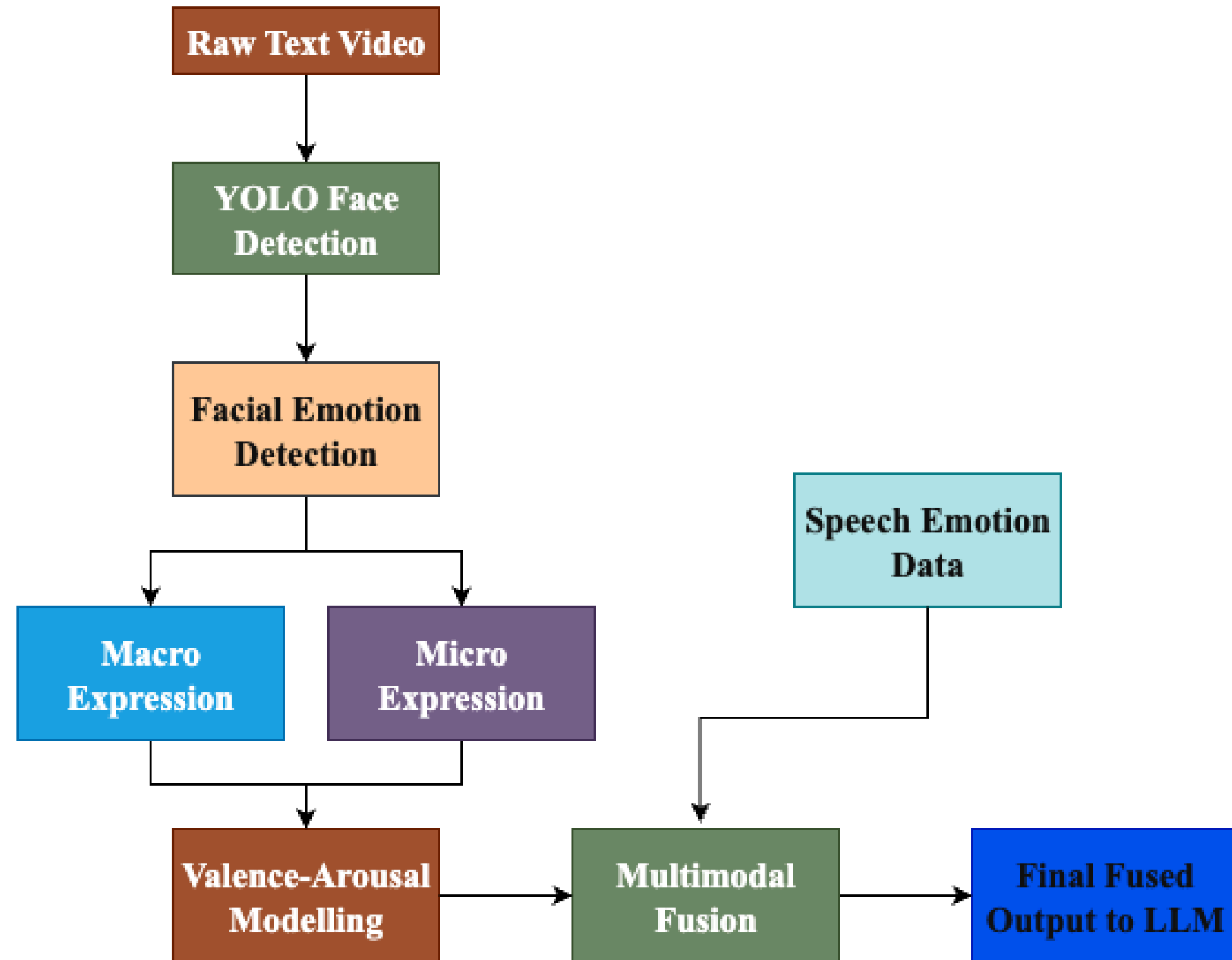
- Empathetic Dialogues (Rashkin et al. 2019): Dataset for empathetic text-based dialogue.
- EDOS (Welivita et al. 2021): Large-scale emotional dialogue dataset with intent labels.
- Multimodal Emotion Recognition (2023–24 works):
CNN/LSTM/transformer models combining facial + speech cues.
- Edge Emotion Recognition (Raspberry Pi prototypes, 2021+): Lightweight CNNs deployed with quantization/pruning.
- LLM Fine-tuning for Empathy (2024–25): Instruction tuning and RLHF for therapy-style safe responses.

PROJECT WORKFLOW

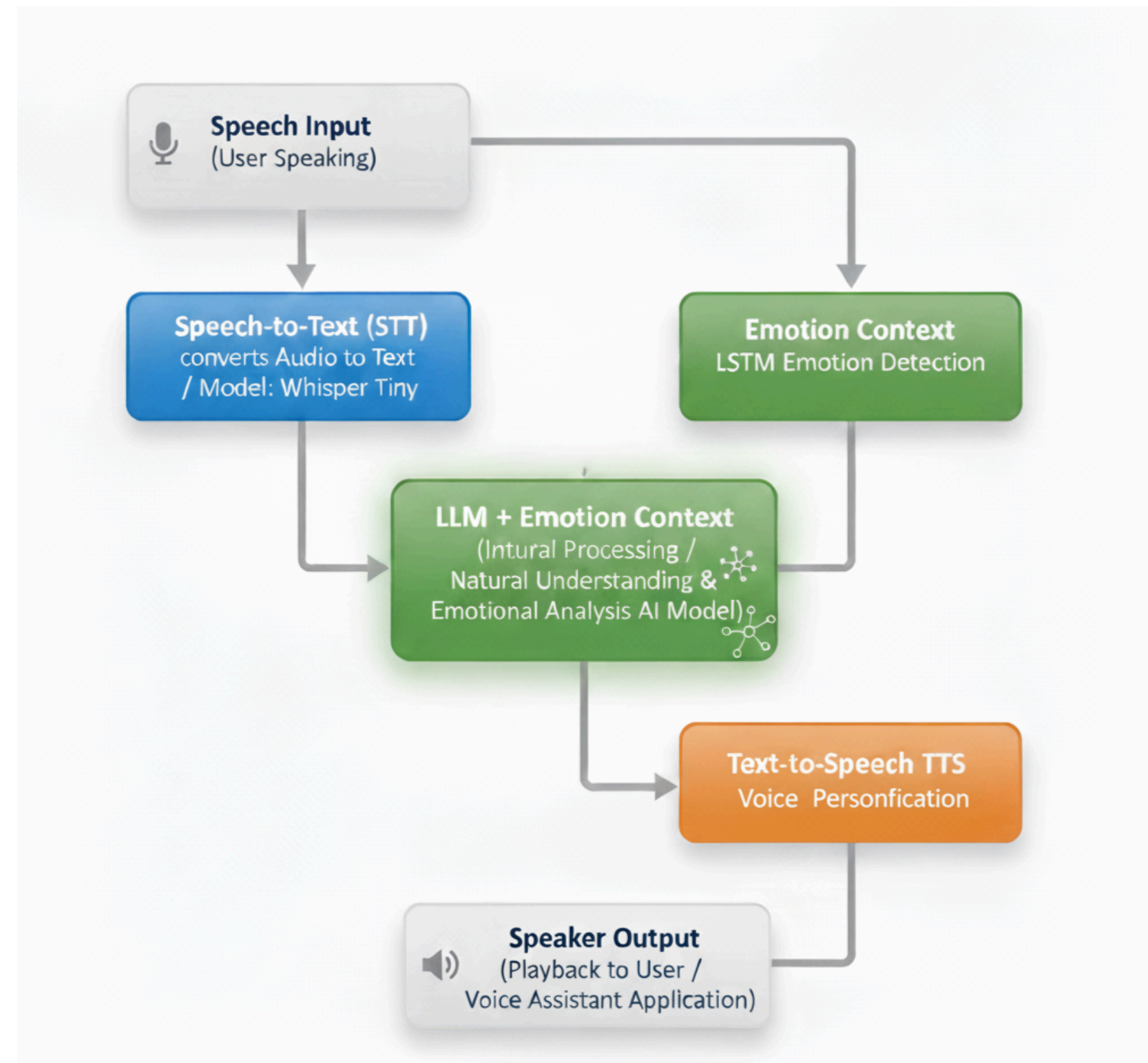


Overall Workflow: Multimodal Sentimental Analysis and LLM-Driven Mental Care

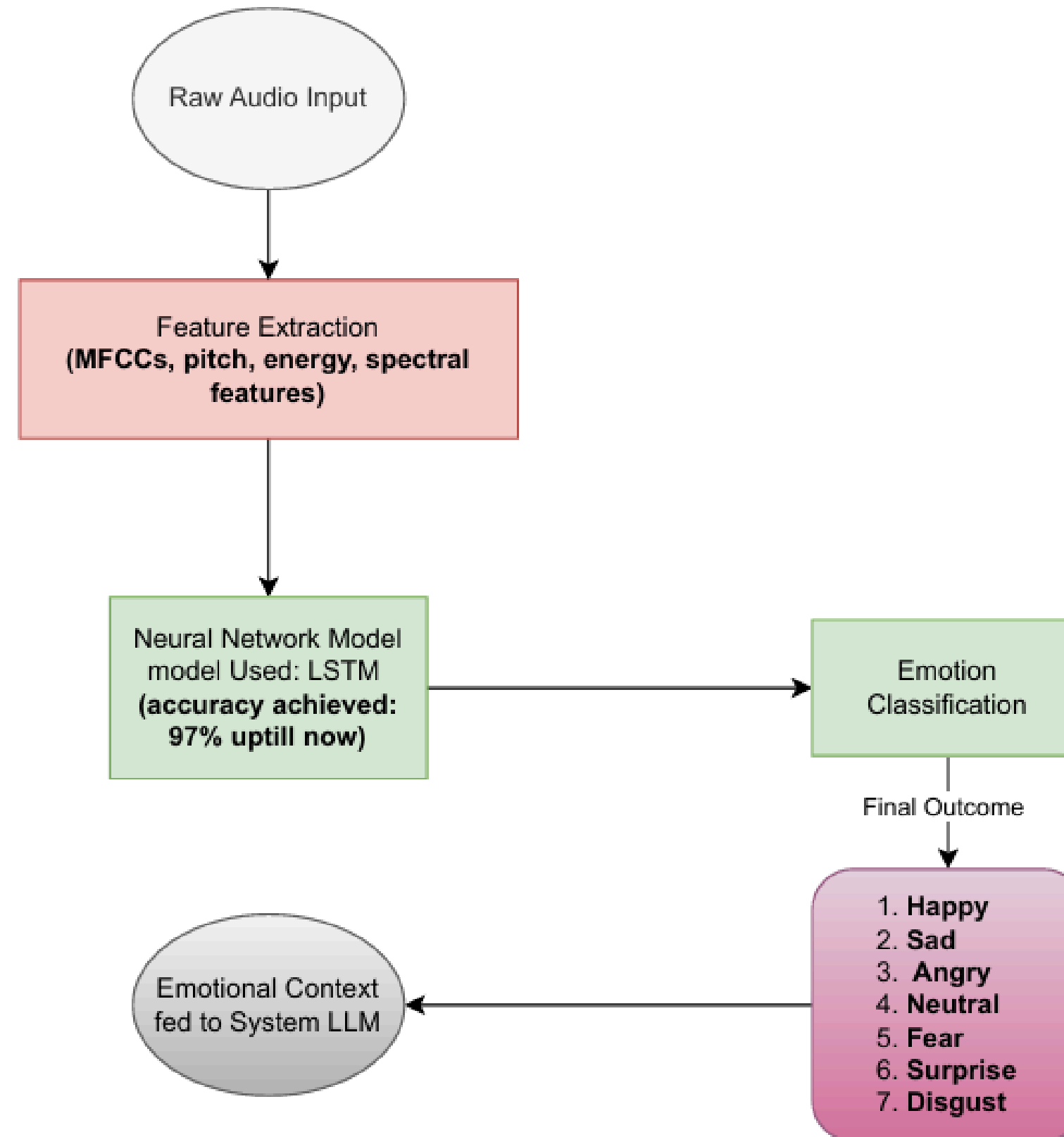
FACIAL RECOGNITION WORKFLOW



SPEECH PROCESSING PIPELINE



SPEECH MODEL FLOWCHART

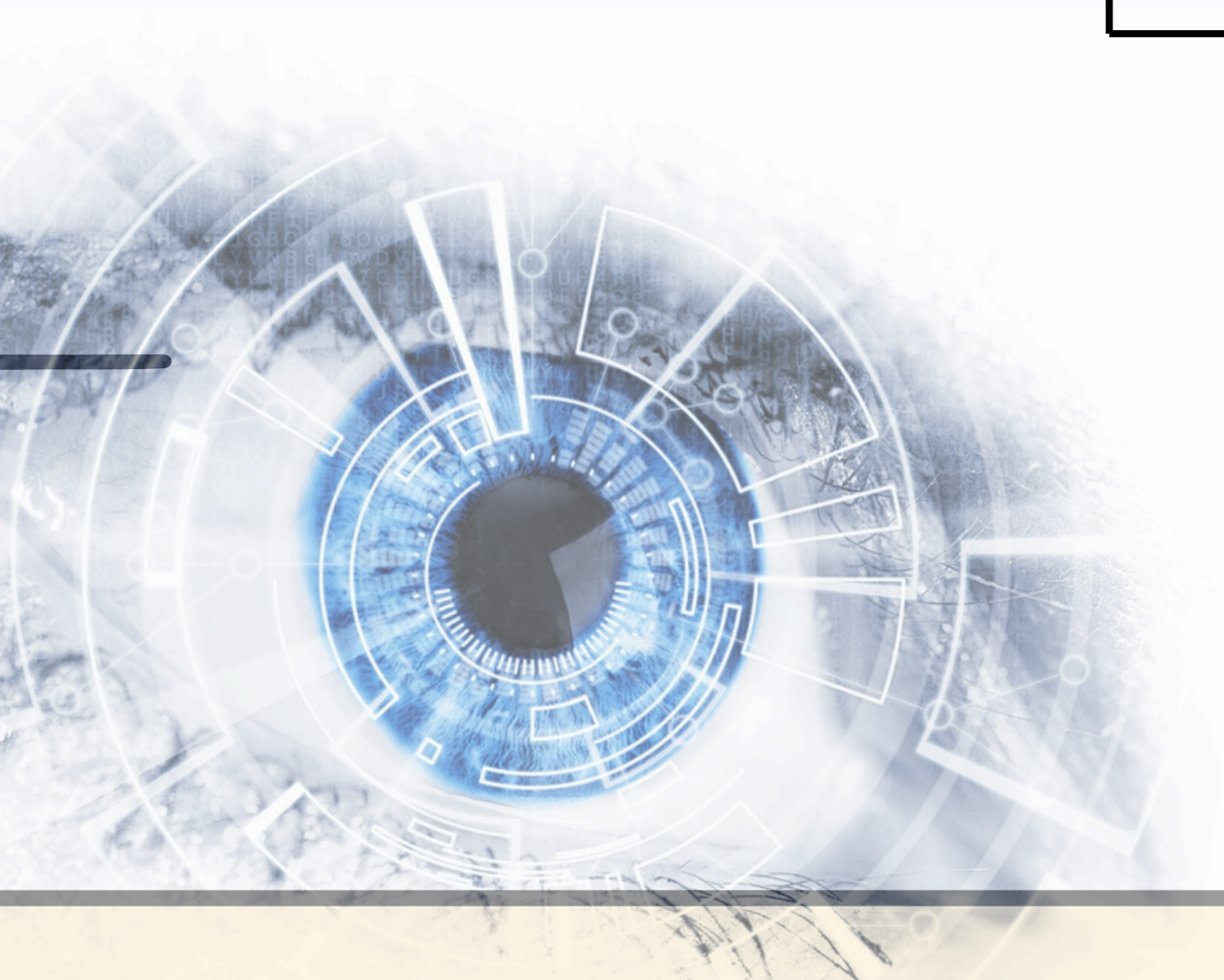


FUSED OUTPUT

Speech Emotion
Detection

Facial Recognition
Output

Final Fused Output
given to System LLM



LLM FINE-TUNING

Base Model: TinyLlama (raw model is unsafe & off-topic)

Goal: Empathetic, psychologist-style dialogue

Steps

- Curate mental-health datasets (therapy scripts, CBT/SFBT prompts)
- Apply instruction fine-tuning + RLHF for safe responses
- Add content-safety filters & ethical guardrails
- Support multilingual conversations

Outcome: Private, on-device mental-health companion that avoids harmful or unethical replies



FINE-TUNING PIPELINE & DEPLOYMENT

Environment

- Hugging Face + PEFT/LoRA
- Train on Kaggle GPU (T4/P100)
- Deploy quantized (INT8/FP16) on Raspberry Pi 5

Process

- Clean & balance therapy datasets (PHQ-9, GAD-7, PCL-5)
- Supervised fine-tune on therapist-client dialogues
- Add RLHF & rule-based safety filters
- Evaluate empathy, coherence, safety

Deployment

- Serve with Ollama locally
- Continuous loop: update → re-fine-tune → redeploy



DATABASE: DESIGN & MANAGEMENT

- Design central emotion & user database.

Stored:

- User profiles & preferences.
- Past emotional states (facial + speech).
- Ensured scalable & organized data handling.
- Importance: Database acts as the “memory backbone,” allowing continuity in therapy sessions.
- Conversation history.



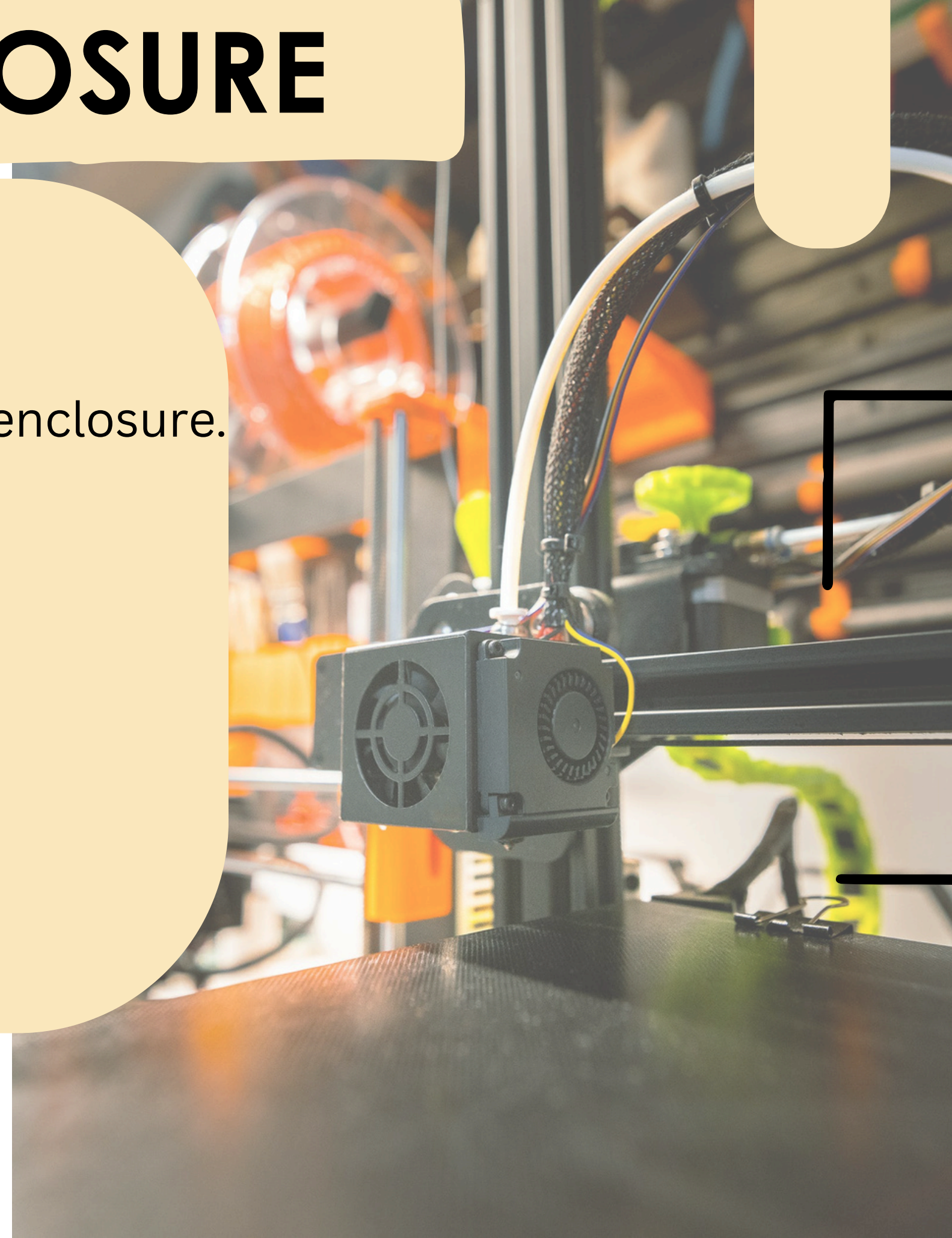
DATABASE-LLM INTEGRATION

- Build efficient linkage between LLM and database.
- Real-time fetching of emotional history.
- Enabling personalized & empathetic responses.
- Reduced repetition in conversations.
- Outcome: Context-aware, human-like interactions instead of generic replies.



DESIGN ENCLOSURE

- Designed & integrated custom Raspberry Pi enclosure.
- Benefits:
- Protected hardware.
- Portable & user-friendly.
- Professional look for deployment.
- Cost-effective and customizable.



DATASETS & QUISTIONNAIRES

1. Emotion Recognition Using CNN:

[FER-2013](#)

2. Speech Recognition using CNN/LSTM:

[RAVDESS Emotional speech audio](#)

3. DAIC-WOZ (Clinical interviews with depression severity labels):

[DAIC-WOZ Database Download - DAIC-WOZ](#).

4. Counselling and Therapy Dialogues:

[EmpatheticDialogues Dataset](#) | [Papers With Code](#)

5. CounselChat Dataset:

[nbertagnolli/counsel-chat · Datasets at Hugging Face](#). && [nbertagnolli/counsel-chat: This repository holds the code for working with data from counselchat.com](#)

6. PHQ-9 Questions, PCL-5 Full List of Questions, GAD-7 Questions

DELIVERABLES AND KPIS

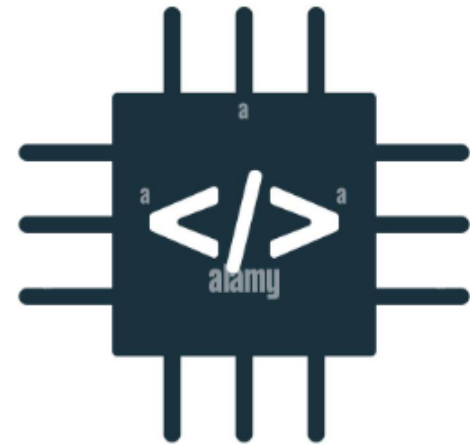


Deliver a real-time multimodal emotion detection and empathetic response system integrating speech, facial emotion recognition, and conversational AI on Raspberry Pi 5.



- Speech Emotion Accuracy: $\geq 90\%$
- Facial Emotion Accuracy: $\geq 90\%$
- Speech-to-Text Accuracy: $\geq 95\%$ (clean audio)
- Response Time: ≤ 3 seconds end-to-end latency
- System Compatibility: Runs in real-time on Raspberry pi with no significant lag

REVELANCE TO COMPUTER ENGINEERING



EMBEDDED SYSTEM

Real-time deployment on Raspberry Pi 5



Speech-to-text, audio preprocessing, voice modulation



Speech & facial emotion recognition,
chatbot, emotional TTS



Database Integration

TEAM STRUCTURE AND RESPONSIBILITIES

Muhammad Taha

- Project management, LLM psychologist fine-tuning, Raspberry Pi 5 deployment and edge optimization/quantization.

Sohaib Afzal

- YOLO face detection, facial emotion (macro/micro) analysis, valence–arousal modeling, multimodal fusion (facial input).

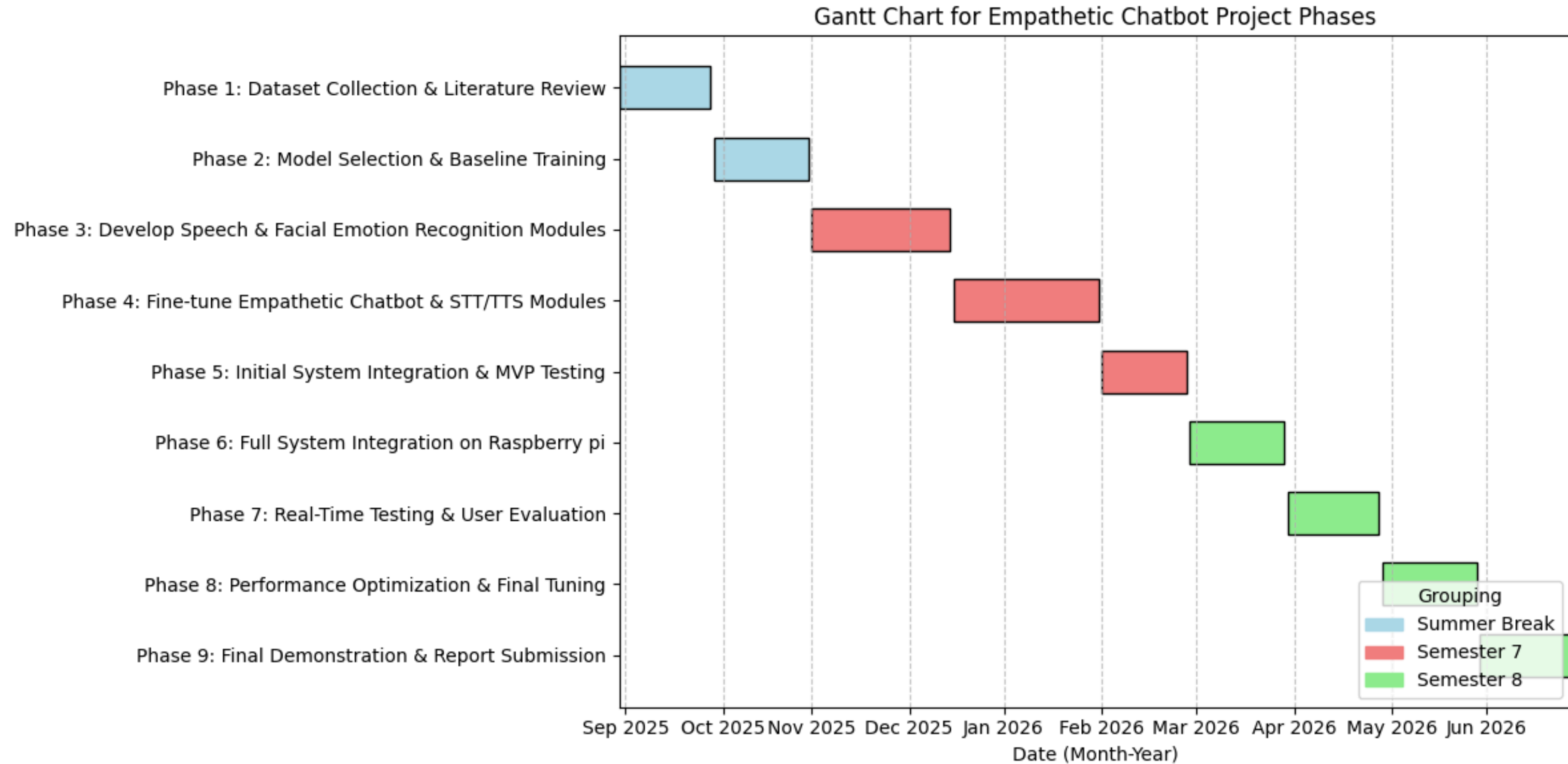
Hashir Ashraf

- Speech-emotion modeling, multilingual STT/TTS refinement, multimodal fusion (audio input), full pipeline integration.

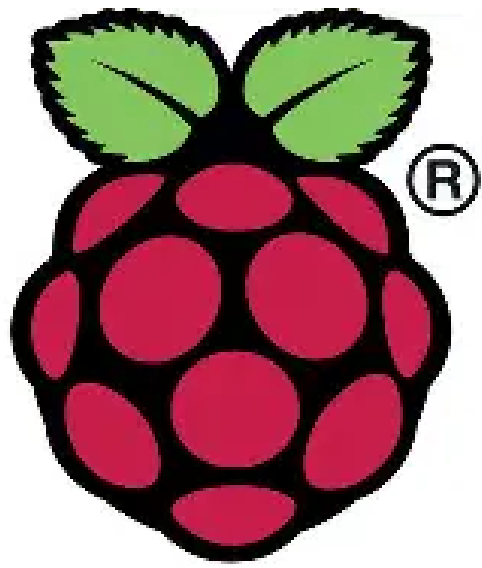
Muhammad Muhtashim

- Co-develop & optimize LLM with Taha, design and manage the central emotion/user database, 3D-printed enclosure integration.

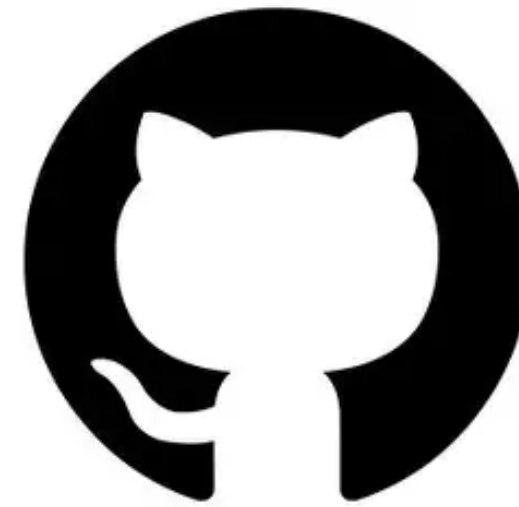
GANTT CHART



PROJECT TOOLS



Raspberry Pi

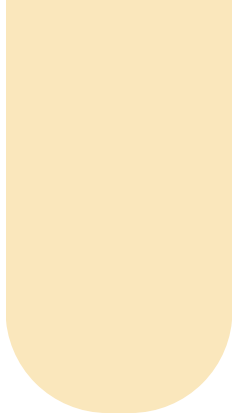
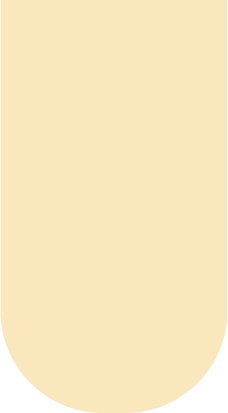


kaggle



CONCLUSION

- LLM Co-development: Enabled real-time, empathetic dialogue tuned for therapy-like sessions.
- Database Architecture: Provided long-term personalization & continuity with emotional history.
- Database-LLM Integration: Allowed context-aware, human-like conversations.
- 3D-Printed Enclosure: Delivered professional, portable, and deployable hardware.
- Overall Impact: A scalable, privacy-preserving, real-time emotional support system.



THANK YOU!



Any Questions?